

Week 5: The Network Layer: Forwarding and Routing, IP Addressing, CIDR, and Subnetting

This report provides a comprehensive analysis of the network layer focusing on its dual architecture forwarding and routing and the evolution of its primary identification system, IPv4 addressing.

I. The Architecture of the Network Layer

The network layer serves as the "narrow waist" of the internet's hourglass architecture, acting as the universal protocol that allows any application to run over any link technology. Its primary goal is to deliver packets from a source host to a destination host across multiple networks.

Data Plane vs. Control Plane

One of the most critical distinctions in networking is the separation of the data plane and the control plane.

- **The Data Plane (Forwarding):** This is the "fast path," operating on timescales of nanoseconds to microseconds. It is local to a single router and involves the hardware-level task of looking at a packet's header and using a Forwarding Information Base (FIB) to decide which output port to use.
- **The Control Plane (Routing):** This is the "slow path," operating on seconds to minutes. It involves network-wide logic, where protocols like OSPF or BGP compute the optimal paths that populate the forwarding tables.

Router Functionality and Design

Routers are specialized devices that connect different networks at Layer 3. Their architecture consists of input ports, a switching fabric, and output ports. When a packet arrives, the router performs several steps: it checks the checksum, decrements the Time to Live (TTL) to prevent infinite loops, and performs a Longest Prefix Match (LPM) to find the correct destination entry in the routing table. To handle high-speed traffic (e.g., 100 Gb/s), routers often utilize TCAM (Ternary Content-Addressable Memory), which allows for parallel searching of all routing table entries in a single clock cycle.

The Shift toward SDN

Traditional networking relies on a distributed control plane where each router independently computes its view of the network. Software-Defined Networking (SDN) challenges this by centralizing the control plane in a software controller that pushes forwarding rules to "dumb" switches. While SDN has revolutionized cloud data centers like those at Google and AWS, it has not replaced the open internet, which still relies on distributed protocols like BGP.

II. IP Addressing and Subnetting

Address management is the second pillar of the network layer, providing the globally unique identifiers needed to name destinations

The Failure of Classful Addressing

Initially, the internet used classful addressing (Classes A, B, and C), which fixed the network and hosted portions of an address at 8, 16, or 24 bits. This system failed due to massive address waste, for example, a company needing 2,000 addresses would be forced to take a Class B block with 65,000 addresses and a resulting routing table explosion

CIDR and Subnetting

To save the internet from address exhaustion, CIDR (Classless Inter-Domain Routing) was introduced in 1993. CIDR allows for flexible prefix lengths (e.g., /21 or /27) and enables aggregation (supernetting), where multiple adjacent blocks collapse into a single routing entry. Subnetting is the process of dividing a large address block into smaller network segments by "borrowing" bits from the host portion to extend the prefix. This creates separate broadcast domains, allowing for better security and organization. Variable-Length Subnet Masking (VLSM) further refines this by allowing subnets within an organization to be of different sizes, though it requires strict bookkeeping to avoid overlapping blocks

Global Allocation and Exhaustion

IPv4 addresses are distributed through a hierarchy: the IANA allocates blocks to Regional Internet Registries (RIRs), which then distribute them to ISPs and organizations. Because there are only ~4.3 billion IPv4 addresses, the free pool has been exhausted across all global regions, with the last being Africa in 2021. The sources note that NAT (Network Address Translation), which allows thousands of private hosts to share a single public IP, is the primary reason the IPv4-based internet has survived despite this exhaustion.

Special Address Ranges

Certain address blocks are reserved for specific uses and are not routable on the public internet:

- **RFC 1918:** Private addresses (e.g., 192.168.x.x, 10.x.x.x) used in homes and offices.
- **Loopback (127.0.0.0/8):** Used for testing local software.
- **Link-local (169.254.0.0/16):** Auto-assigned when a DHCP server cannot be reached

Conclusion

Together, these lectures establish the fundamental "how" and "where" of the network layer. Lecture 13 defines the mechanical architecture of routers and the division of labor between data and control planes, while Lecture 14 provides the mathematical foundation for naming and

dividing the network space. This sets the stage for upcoming topics such as specific routing algorithms (Dijkstra and Bellman-Ford) and the transition to IPv6

Personal Insights & Reflections:

Beyond the technical formulas, these lectures completely shifted my perspective on how the internet functions. I never thought about the internet to me it was just something that happened and these two lecture there so eye opening,

The "Best-Effort": I always assumed the internet was a perfectly reliable machine. Learning that IP operates as a "best-effort" service where data delivery is never guaranteed was the most surprising part. The End-to-End Argument makes perfect sense now: forcing routers to verify every packet wastes computational power when the endpoints (the apps) have to do those checks anyway.

The "Good Enough" Principle: It was a reality check to learn that the internet isn't the "best" engineered option, it's just the one that won. Systems like ATM or X.25 were "perfectly engineered" but lost because IP was cheaper, simpler, and ran on commodity hardware. It shows that in the real world, scalability and cost beat "perfect" engineering every time . (This was valuable even for the real-world engineering work and it really stuck with me)

The Scarcity Reality: I was shocked to learn we actually ran out of free IPv4 addresses in 2011. Seeing how corporate "land grabs" in the 80s (like Ford and Apple getting massive /8 blocks) led to our current exhaustion highlights how unpredictable the scale of the internet was to its creators.

Engineering in the Wild: Learning that early internet security was neglected because the original engineers "all knew each other" was a fascinating look at the human side of tech. It's a reminder that the world's most complex system was built in a somewhat chaotic, high-trust environment that we are still retrofitting for security today.